



DAAN862: ANALYTICS PROGRAMMING IN PYTHON

Lesson 3: Statistical Analysis with Pandas

Lesson 3.2: Basic Data Types: Data Frame

Another basic data type in Python is data frame, which is equivalent to the dat frames in R. A **DataFrame** is a table of data which use columns and rows to represent variables and instances, respectively. The DataFrame has both a column and row index.

(click the tabs to learn more)

dict

Sequence

Numpy Array

Creating a DataFrame with dict

In Figure 3.9, data is dict which has three keys ['state', 'year', and 'pop'] and three lists as their values. When you create a DataFrame from this dict, it will use dict keys as column names and dict values as elements.

```
In [19]: data = {'state': ['Ohio', 'Ohio', 'Ohio', 'Nevada', 'Nevada'],
...:            'year': [2000, 2001, 2002, 2001, 2002],
...:            'pop': [1.5, 1.7, 3.6, 2.4, 2.9]}
```

```
In [20]: frame = DataFrame(data)
```

```
In [21]: frame
```

```
Out[21]:
   state  year  pop
0  Ohio  2000  1.5
1  Ohio  2001  1.7
```

